

Christopher Musco

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Education

Massachusetts Institute of Technology

Cambridge, MA

Ph.D. Candidate, Computer Science – Theory of Computation

2013 – present

Advisors: Jonathan Kelner, Martin Rinard

Yale University

New Haven, CT

B.S. Applied Mathematics, B.S. Computer Science

2008 – 2012

AM Thesis: Fast Discrete Laplace Transforms (under Vladimir Rokhlin)

CS Thesis: Graph Constructions for Machine Learning (under Daniel Spielman)

Research Interests

I am interested in dimensionality reduction and its application to randomized numerical linear algebra, sketching and streaming algorithms, and machine learning. I also work on problems related to algorithmic graph theory and collaborate with MIT's Center for Resilient Software on quantitative approaches to computer security and approximate computation.

Academic Positions

Massachusetts Institute of Technology

2013 – present

Research Assistant

Teaching and Mentorship

Research Science Institute/MIT

2014

Summer Research Advisor

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program.

- Designed a project on concurrent data structures under the direction of Professor Martin Rinard.
- Chris' paper, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Yale University

Teaching Assistant, Introduction to Computer Science

2012

Covered functional programming, boolean algebra, formal language theory, and basic complexity theory.

Teaching Assistant, Introduction to Programming

2011

Led weekly section for 15-25 students covering basic Java programming. Tutored 5-6 students individually.

Undergraduate Grader, Introductory Calculus

2009 – 2010

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

Honors and Awards

National Science Foundation

2014

Graduate Research Fellowship

Yale University

2012

Cum Laude, Distinction in Both Majors

Publications

Conference Papers.....

[Dimensionality Reduction for K-Means Clustering and Low Rank Approximation](#). Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *To appear in STOC* 2015.

[Principled Sampling for Anomaly Detection](#). Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiroglou-Douskos, Martin Rinard. *NDSS* 2015.

[Uniform Sampling for Matrix Approximation](#). Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *ITCS* 2015.

[Single Pass Spectral Sparsification in Dynamic Streams](#). Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *FOCS* 2014. (Invited to SICOMP Special Issue).

Other.....

[Stronger Approximate Singular Value Decomposition via the Block Lanczos and Power Methods](#). Cameron Musco, Christopher Musco. *Preprint* 2015.

Talks

Dimensionality Reduction for K-Means Clustering

MIT, Theory Lunch

June 2014

Principled Sampling for Anomaly Detection

MIT, Defense Advanced Research Projects Agency Site Visit

Mar. 2015

Network and Distributed System Security Symposium (NDSS)

Feb. 2015

Single Pass Spectral Sparsification in Dynamic Streams

MIT, Annual Sublinear Algorithms Day (Poster Session)

Apr. 2015

Harvard University, TOC Seminar

Nov. 2014

Foundations of Computer Science (FOCS)

Oct. 2014

Service

External Reviewer.....

European Symposia on Algorithms (ESA)

2015

Foundations of Computer Science (FOCS)

2015

International Colloquium on Automata, Languages and Programming (ICALP)

2015

Network and Distributed System Security Symposium (NDSS)

2015

Professional Experience

Redfin

Seattle, WA

Software Engineer, Data Team

2012 – 2014

Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code base.

Elysium Digital

Cambridge, MA

Summer Consultant

Summer 2011

Provided technical expertise in software patent cases at leading litigation consulting company.

Amicus*Software Developer*

Built tools for political campaign management and social fundraising at early-stage startup.

New Haven, CT*2010 – 2011***Moody's Investors Service***Intern, Asset Finance Group*

Established and monitored bond ratings on asset securitizations.

New York, NY*Summer 2009***U.S. Department of Agriculture***Intern, Natural Resources Conservation Service*

Collected and analyzed data for state conservation and research projects.

East Greenwich, RI*Summer 2007*